

Veronica Wang

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EDUCATION

Stanford University

B.S. Computer Science

Stanford, CA

June 2029

Coursework: Machine Learning, Computer Vision, Applied Matrix Theory, Computer Organization & Systems, Linear Algebra & Multivariable Calculus, Trust and Safety, Probability for CS, Programming Abstractions

SKILLS

Python, Java, C++, C, Onshape, Fusion 360, Vercel, Git, Unix, GCP, AWS, Modal, React, Next.js, Node.js, Figma, Unity, Arduino, Pytorch, SciPy, NumPy, Scikit-learn, x86, Supabase, MongoDB, Docker, Mathematica, LaTeX, Excel

EXPERIENCE

Boeing | Systems Software Engineer

Incoming June 2026

- Low-level mission systems software and virtual integration for the 737 Next Gen/Poseidon P-8 aircraft.

University of Maryland | Quantum Research Intern

May-August 2024

- Co-authored paper accepted to European Physical Journal A - "Hadrons and Nuclei" regarding evidence for the computational cost of adiabatic quantum state preparation.
- Built an automated simulation pipeline with Python to orchestrate 1k+ runs with strict $\leq 10\%$ error gates on UMD's supercomputing cluster to test the novel conjecture about quantum simulation of field theories.
- Used Mathematica and found a 97% improvement in efficiency versus standard practice and found that cost scales superlinearly with path length.

Chimes | Assistive Technology Designer

October 2023-Jan 2024

- Designed an RFID-based audio identification device to be scaled to hundreds of visually impaired employees at Chimes, a nonprofit employing people with disabilities, nationwide. Device helps custodial workers distinguish cleaning supplies without relying on sight or tactile cues.
- Built a two-ESP32 pipeline with RFID reader, ESP-NOW transmission, and SD card audio playback through a CAD-modeled 3D-printed enclosure.
- Blindfolded user study across 36 trials showed 30.1% faster item identification.

PROJECTS

Transformer-based Neurosurgical Phase Classification

- Partnered with Stanford School of Medicine for dataset access
- Designed NeuroOperA, a causal Transformer for phase recognition on annotated aneurysm clipping videos (40,725 labeled frames across 4 surgical phases). Model outperformed the prior OperA architecture with Viterbi segmental F1 at 10 of 0.944 vs. ~ 0.80 , and achieved $\sim 95\%$ validation accuracy vs. OperA's 92%.
- Showed that fine-tuning ResNet50 on surgical frames (vs. frozen ImageNet weights) was the single largest factor: frame accuracy 53% $\rightarrow$ 95.7%, F1 at 10 0.553 $\rightarrow$ 0.895. Applied Viterbi decoding with a learned transition matrix, outperforming hand-crafted surgical priors across all thresholds

JobShield: A Full-Stack Fraudulent Job Posting Detection Platform

- Built a hybrid fraud-detection platform targeting active malware attack vectors (OtterCookie) using Next.js, Supabase, and Gemini 2.5 Flash. Achieved F1 0.913 and 95% fraud recall at 4.5x lower inference cost than a pure-LLM baseline (\$0.077/1k vs. \$0.35/1k).
- Logistic Regression + TF-IDF on posting text with 8 metadata flags handles obvious cases in under 0.05ms. Automatic feedback loop injects every moderator decision as a labeled few-shot example into subsequent LLM calls for continuous improvement without retraining.

Illicit Bitcoin Wallet Classification

- Proposed EvolveGCN-T, replacing GRU-based weight evolution with a Transformer encoder self-attending over GCN weight matrix history. Outperformed EvolveGCN-O on Bitcoin-OTC edge classification (micro-F1 0.783 vs. 0.699, +8.4 pts) and reproduced published baselines to within $\pm 1\%$.
- Designed a leakage-free evaluation protocol for dynamic graph neural networks by removing train-test graph overlap, improving robustness and generalization analysis for financial fraud detection.

Awards

American Invitational Math Examination (AIME) Qualifier, USA Computing Olympiad (USACO) Gold Qualifier, NCWIT Aspirations in Computing Maryland Winner